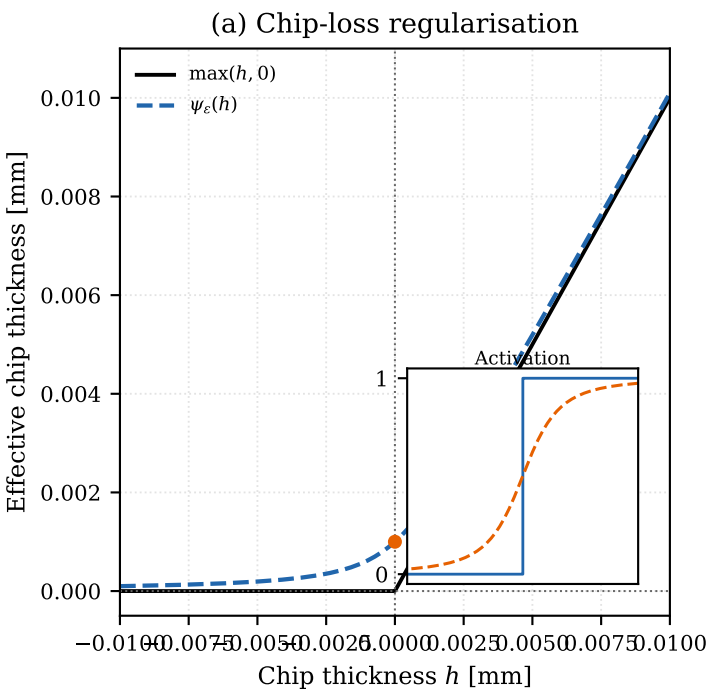
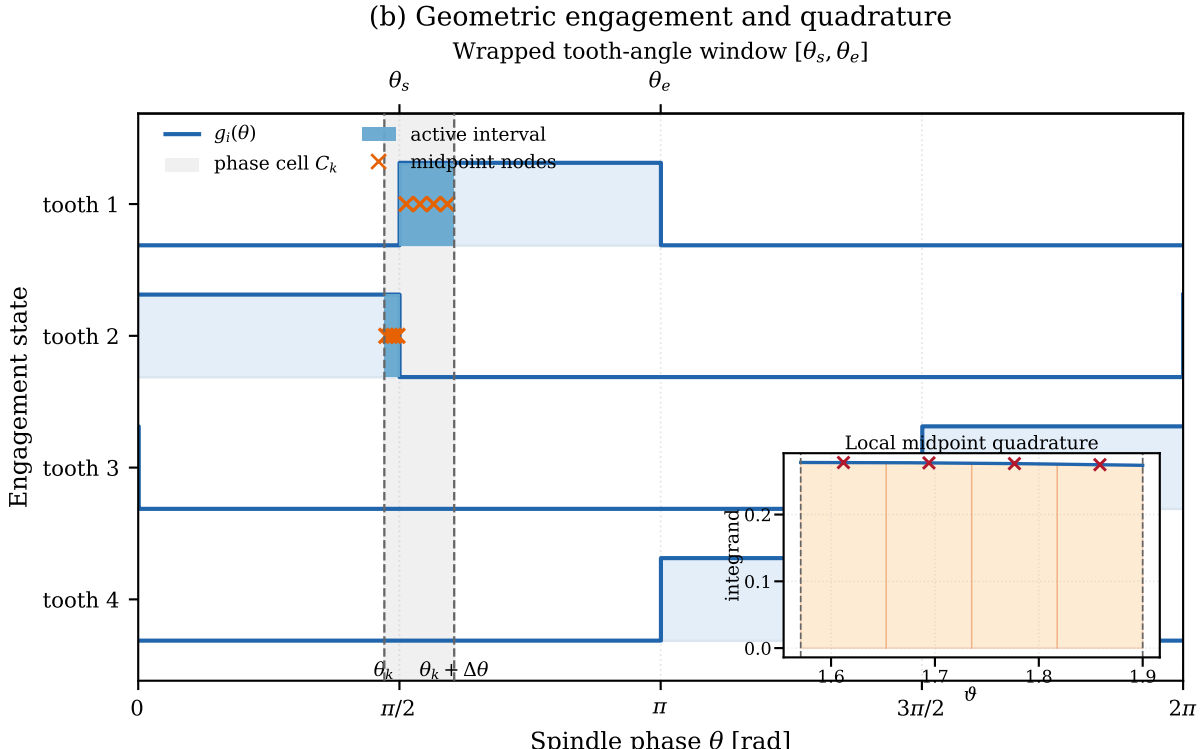

(a) Chip-loss regularisation
max(h, 0)
ψε(h)
Activation
1
0
Effective chip thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
0.0100
Chip thickness h [mm]


(b) Geometric engagement and quadrature
Wrapped tooth-angle window [θs, θe]
θs
θe
gi(θ)
active interval
phase cell Ck
midpoint nodes
tooth 1
tooth 2
tooth 3
tooth 4
Engagement state
Local midpoint quadrature
integrand
0.2
0.1
0.0
1.6
1.7
1.8
1.9
ϑ
θk
θk + Δθ
0
π/2
π
3π/2
2π
Spindle phase θ [rad]